Nonnegative Matrix Functional Factorization for Hyperspectral Unmixing With Nonuniform Spectral Sampling

Ting Wang, Jizhou Li, *Member, IEEE*, Michael K. Ng, *Senior Member, IEEE*, and Chao Wang

***Abstract*— Unmixing is a crucial technique in analyzing hyperspectral imaging (HSI) data, which involves identifying the endmembers present in the data and estimating their abundance maps. Due to some practical constraints in the atmospheric environment, HSI data is usually nonuniformly distributed along the spectral domain, which brings incomplete spectral information in the hyperspectral unmixing. To overcome this issue, we propose, in this article, nonnegative matrix functional factorization (NMFF) which is an extension of classical nonnegative matrix factorization (NMF) for hyperspectral unmixing. In particular, we present a novel functional factorization model by incorporating the implicit neural representations (INRs) to learn about endmembers. Our method effectively characterizes endmembers by learning a continuous representation through INR with positional encoding, capturing the nonuniform distribution of spectral wavelengths. This distinct approach streamlines NMFF's iterative process for abundance extraction, bypassing the conventionally complex and cumbersome processing. When tested on various datasets, our hyperspectral unmixing approach consistently outperforms established techniques, showcasing the enhanced capabilities of our proposed model.**

***Index Terms*— Hyperspectral unmixing, implicit neural representation (INR), nonuniform sampling, nonnegative matrix factorization (NMF), positional encoding, spectral domain.**

I. Introduction

HYPERSPECTRAL imaging (HSI) data, renowned for their capacity to capture extensive information across multiple spectral bands, have found widespread use in various fields, including environmental monitoring [1], agriculture [2], remote sensing [3], mineral exploration [4], and beyond. The abundance of spectral information within hyperspectral data provides unparalleled insights into material composition, facilitating detailed analyses such as image restoration [5], classification [6], and the detection of objects or anomalies [7]. Amid this versatility, one of the most prominent and intricate problems in hyperspectral data processing is spectral unmixing. It entails the separation of mixed pixel spectra into their fundamental pure spectral signatures, known as endmembers, along with their corresponding abundance fractions. Spectral unmixing is essential for enhancing the interpretability of hyperspectral imagery, facilitating material identification, and enabling quantitative analysis. Spectral unmixing becomes considerably more complex when the endmembers are also unknown, a scenario referred to as spectral blind unmixing. A range of methods have been proposed for spectral blind unmixing, broadly classified into two main categories: physics-based and data-driven.

Physics-based techniques leverage statistical properties to estimate light scattering behavior and interactions among various materials present in the imaged scene. Noteworthy statistical algorithms encompass nonnegative matrix factorization (NMF) [8], [9], independent component analysis (ICA) [10], and Bayesian approaches [11]. Among these, NMF stands out as a suitable choice for hyperspectral unmixing due to its inherent nonnegative outputs and interpretability. Consequently, numerous methodologies based on NMF have been developed to achieve enhanced unmixing performance, as indicated by various studies [12], [13].

On the other hand, many researchers have concentrated on hyperspectral unmixing from a data-driven perspective [14]. The main idea is to leverage autoencoders kind of neural network to achieve more competitive results in unmixing. These networks are fed with reflectance spectra as inputs, generating reconstructed spectra as outputs. The encoder converts the input spectra into fractional abundances, while the decoder, utilizing linear layers with endmembers as weights, transforms these abundances into the reconstructed spectra. Notable instances of such unmixing methodologies include Autoencoder [15], CNNAEU [16], EndNet [17], SNSA [18], DAEN [19], DeepGUn [20], and uDAS [21]. In addition, several convolutional autoencoder-centered unmixing method-

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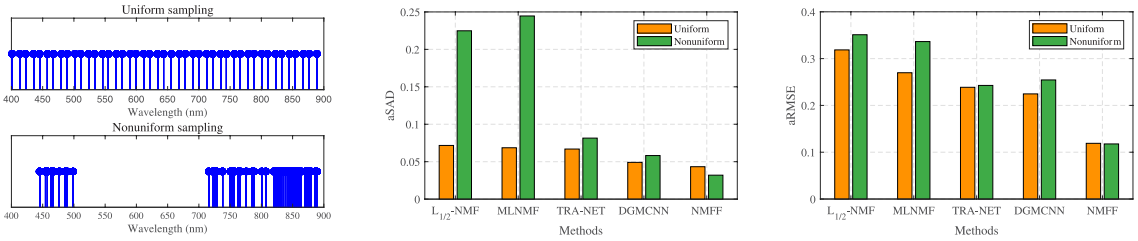


Fig. 1. (Left) Uniform and nonuniform sampled 45 bands corresponding to wavelength with Samson dataset. (Middle) aSAD for endmembers of different typical algorithms are evaluated from the uniform and nonuniform sampling. (Right) aRMSE for abundances of different typical algorithms are evaluated from the uniform and nonuniform sampling. The aSAD and aRMSE of all methods with nonuniform sampling were higher than those with uniform sampling.

ologies have emerged to integrate spatial coherence among adjacent pixels effectively. UnDIP [22] utilized a simplex volume maximization (SiVM) technique to extract endmembers. TRA-NET [23] introduced a new unmixing strategy that combines a convolutional autoencoder with a transformer architecture. Additionally, DGM [24] proposed a comprehensive two-stream deep learning-based framework designed for spectral unmixing. This framework tackles the challenge of endmember variability by yielding pixel-wise endmembers derived from their latent representations. For instance, existing methods often necessitate a substantial number of training samples, lack interpretability in nonlinear unmixing mechanisms, and encounter challenges in estimating endmembers situated in diverse locations within the scene.

Due to practical constraints, such as sensor characteristics, atmospheric effects, and limited resources, hyperspectral image sensors' spectral measurements are not uniformly distributed across the observed scene [25], [26]. This nonuniform sampling pattern can lead to information loss and uneven representation of spectral characteristics. As shown in Fig. 1, we conducted a comparative analysis of the unmixing performance of different methods using the Samson dataset under both uniform and nonuniform wavelength distributions. The results indicate that some existing methods, especially model-based approaches, exhibit inferior unmixing performance when the wavelengths are distributed nonuniformly. So developing advanced methods tailored specifically for nonuniformly sampled hyperspectral data analysis is essential to address these challenges and achieve accurate and reliable hyperspectral data. In our work, we explore how spectral wavelengths can continuously represent the endmembers even under the circumstance of nonuniform sampling wavelengths. The core of our proposed method is based on implicit neural representation (INR) [27], [28], [29] to represent endmembers' information. When faced with nonuniform distribution wavelength configurations, INR can flexibly learn the corresponding endmembers at specific wavelength or band locations, demonstrating its robustness to nonuniform sampling wavelength patterns. By accurately estimating the endmembers, we utilize the iterative framework of NMF to assess abundance. Importantly, this intuitive concept can be formulated into an optimization framework that aligns with NMF and functional approximation principles. Rigorous experiments substantiate that the proposed algorithm yields outstanding unmixing results. By testing three real datasets, our method improves the NMF-based method by up to 68.47% and 63.55% in terms of the average root mean square error (aRMSE) for abundance and the aSAD for the endmember, respectively. Furthermore, our model only utilizes a very simple architecture of a neural network to solve endmember and does not involve any regularization in estimating the abundance. We outperform TRA-NET, a recent work built by the transformer, by up to 53.04% concerning aRMSE.

The main contributions are summarized as follows.

1) We introduce a novel learning-based NMF architecture designed to address the uneven representation of spectral features, thereby enhancing unmixing accuracy. The proposed model operates in an unsupervised manner and is well-suited for remote-sensing data where ground-truth (GT) might be unavailable.
2) For the first time, we integrate an INR to model endmembers. This sets the stage for abundance estimation to be effectively handled within an NMF optimization context.
3) We conduct quantitative and qualitative assessments of the proposed algorithm using datasets with uniform and nonuniform sampling patterns. Our evaluation demonstrates that the algorithm shows favorable performance compared to existing methods, particularly in terms of accuracy in estimating endmembers and abundances.

The remainder of this article is organized as follows. Section II introduces the related works with INR and NMF unmixing approaches. Section III presents the proposed model and its implementation detail. Experimental results and ablation discussion are presented in Section IV. Finally, Section V draws a conclusion and future outlook.

II. Related Works

A. Implicit Neural Representation

While conventional methods rely on grid-like structures for images, discrete amplitude samples for audio, or voxel-based descriptions for 3-D shapes, INR takes a different route by representing signals as continuous functions. These enable mapping signal domains (e.g., pixel coordinates in images) to arbitrary values (e.g., RGB values). Although these functions typically lack straightforward analytical solutions, INRs leverage deep neural networks, specifically multilayer perceptrons (MLPs), to approximate these implicit functions. This approach generates highly detailed and realistic reconstructions, overcoming the limitations of explicit mathematical representations for complex natural image structures.

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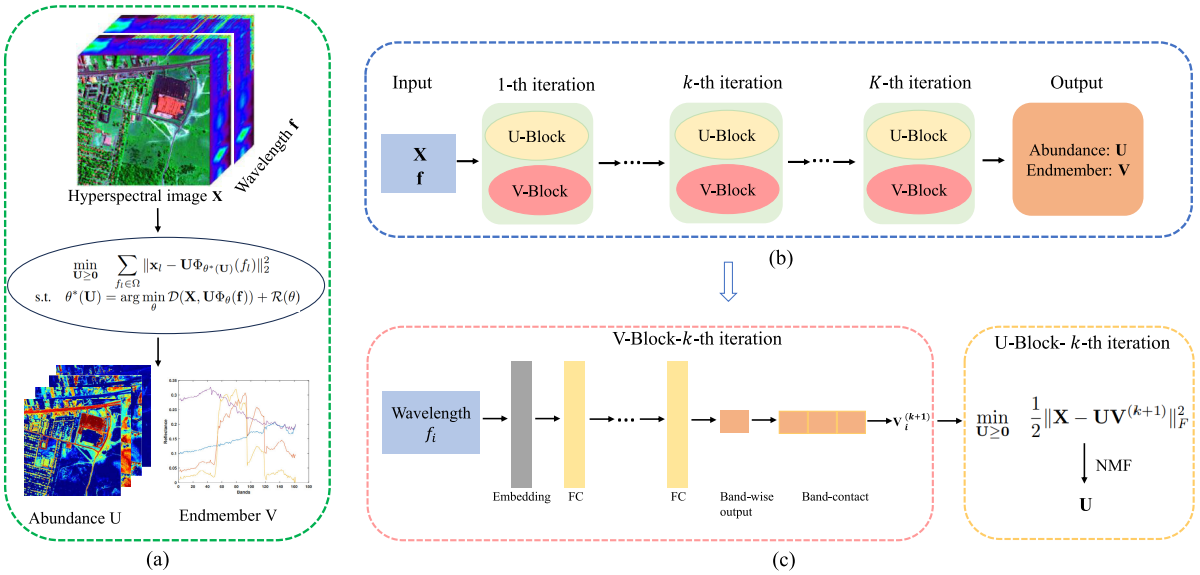


Fig. 2. Flowchart of the proposed algorithm. (a) Procedure of the NMFF and optimized conventional iterative algorithm for estimating the endmember and abundance maps. (b) Outline of the framework's architecture. (c) Operational steps for the kth layer within the U-Block and V-Block.

INR has found extensive applications in computer vision, with numerous works proposing its use in various 3-D vision tasks [28], [30]. Recent works have explored its potential for videos [31], [32] and 2-D images [28], [33]. However, most existing methods are primarily designed for natural images or videos. Most recently, [34], [35] used the INR for HSI super-resolution, exhibiting promising and attractive properties. In these approaches, MLPs were frequently used to approximate INRs, taking spatial or spatiotemporal coordinates as input and outputting the signals at that specific point. In the context of hyperspectral images, our method utilizes a purposely designed MLP that takes the corresponding wavelength in the band as input and directly outputs the endmember values of that specific spectral band location.

B. NMF-Based Unmixing Models

NMF-based methods are based on the linear mixing model (LMM) which assumes an observed pixel spectrum in an HSI can be produced by a linear combination of endmember signatures and their corresponding abundances. Let $\mathbf{X} \in \mathbb{R}^{N\times L}$ denote an HSI with $N = n \times m$ pixels, L spectral bands. The matrix formulation of LMM can be described as

$$\mathbf{X} = \mathbf{UV} + \mathbf{E} \tag{1}$$

where $\mathbf{V} \in \mathbb{R}^{J\times L}$ denotes endmember matrix, $\mathbf{U} \in \mathbb{R}^{N\times J}$ represents the abundance matrix for all endmembers, and $\mathbf{E} \in \mathbb{R}^{N\times L}$ represents the noise matrix, J is the number of the underlying endmembers. Typically, two constraints are imposed on $\mathbf{U}$, i.e., the abundance of nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC), given by $\mathbf{U} \geq \mathbf{0}$ and $\mathbf{U}\mathbf{1}_J = \mathbf{1}_N$, here $\mathbf{1}_J$ and $\mathbf{1}_N$ are all-one column vectors with size J and N, respectively. NMF [36] is primarily concerned with breaking down $\mathbf{X}$ into the product of two nonnegative matrices, namely $\mathbf{U}$ and $\mathbf{V}$. In other words, it aims to approximate $\mathbf{X}$ as $\mathbf{UV}$ where their combined pixels of $\mathbf{UV}$ are nonnegative. This particular factorization aligns with the LMM, making NMF an appealing approach for hyperspectral unmixing. Generally, the Frobenius norm is employed to quantify the degree of approximation between $\mathbf{X}$ and $\mathbf{UV}$. For the implementation of NMF, a crucial consideration revolves around initializing the associated variables. The endmember matrix $\mathbf{V}$ can be initialized by various methods such as random values from 0 to 1, vertex component analysis (VCA) [37]. Abundance matrix $\mathbf{U}$ is initialized according to the fully constrained least squares (FCLSs) algorithm [38].

III. Methodology

In this section, we introduce the proposed nonnegative matrix functional factorization (NMFF) for HSI unmixing, as illustrated in Fig. 2. Our approach consists of two iterative blocks: the abundance ($\mathbf{U}$-block) and the endmember ($\mathbf{V}$-block). The endmember block parameterized by neural networks generates results by referencing the band index exclusively. The abundance block iteratively solves for matrix $\mathbf{U}$ without requiring any regularization terms.

A. Motivation

HSI unmixing becomes increasingly challenging when there is an uneven distribution of wavelengths across the electromagnetic spectrum. This unevenness can hinder the accurate identification of the contributing endmembers [39]. Following the NMF framework, we consider the optimization model:

$$\min_{\mathbf{U},\mathbf{V}\geq\mathbf{0}} \quad \frac{1}{2}\|\mathbf{X} - \mathbf{UV}\|_F^2 \tag{2}$$

where the operator $\|\cdot\|_F$ denotes the Frobenius norm. We denote the endmembers as $\mathbf{V} \in \mathbb{R}^{J\times L}$. Exploring endmember representations that align with the continuous variation of frequencies is essential to address this challenge and enhance accuracy. We define the endmembers with smooth and continuous relationships to frequencies as

$$\mathbf{V} = \left[\Phi_\theta(f_1), \Phi_\theta(f_2), \ldots, \Phi_\theta(f_L)\right] \tag{3}$$

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where f_l is the wavelength in the lth band, a continuous representation $\Phi_\theta(\cdot) : \mathbb{R} \to \mathbb{R}^J$ contains the parameter θ. This approach aims to leverage the inherent patterns in spectral data, promising more robust and precise spectral unmixing results. In essence, the motivation behind this approach is to overcome the complexities arising from wavelength nonuniformity and achieve superior accuracy and effectiveness in hyperspectral unmixing.

B. Model Formulation

Given an HSI data $\mathbf{X}$, we reformulate the LMM model (1) under a continuous representation

$$\mathbf{X} = \mathbf{U}\Phi_\theta(\mathbf{f}) + \mathbf{E} \tag{4}$$

where $\Phi_\theta(\mathbf{f}) = [\Phi_\theta(f_1), \Phi_\theta(f_2), \ldots, \Phi_\theta(f_L)]$ is a J-by-L matrix from the continuous representation Φ_θ. Here, we use a neural network to represent Φ_θ. Similar to (2), we formulate the spectral unmixing into the following bilevel optimization problem:

$$\begin{aligned} \min_{\mathbf{U}\geq\mathbf{0}} \quad & \sum_{f_l\in\Omega} \|\mathbf{x}_l - \mathbf{U}\Phi_{\theta^*(\mathbf{U})}(f_l)\|_2^2 \\ \text{s.t.} \quad & \theta^*(\mathbf{U}) = \arg\min_\theta \mathcal{D}(\mathbf{X}, \mathbf{U}\Phi_\theta(\mathbf{f})) + \mathcal{R}(\theta) \end{aligned} \tag{5}$$

where $\Omega = \{f_1, \ldots, f_L\}$ is the set of wavelength on the available bands. Here, the upper-level optimization subproblem is the optimization problem for $\mathbf{U}$, and the lower-level optimization subproblem pertains to the optimization problem regarding θ. Its objective is to find the optimal $\mathbf{U}$ given the lower-level optimization subproblem for $\theta^\star$ and to find the optimal $\theta^\star$ given the upper-level optimization subproblem for $\mathbf{U}$. Here $\mathcal{D}$ is the data-fidelity term and $\mathcal{R}$ is the regularization. Specifically, we employ an MLP with parameters θ to parameterize Φ. Our model involves a factorization between a nonnegative matrix $\mathbf{U}$ and a continuous function Φ. Hence, we call it the NMFF.

To solve the minimization problem (5), we can alternatively update $\mathbf{U}$ and the MLP. That is to say

$$\begin{cases} \theta^{(k+1)} = \arg\min_\theta \mathcal{D}\big(\mathbf{X}, \mathbf{U}^{(k)}\Phi_\theta(\mathbf{f})\big) + \mathcal{R}(\theta), \\ \mathbf{U}^{(k+1)} = \arg\min_{\mathbf{U}\geq\mathbf{0}} \sum_{f_l\in\Omega} \|\mathbf{x}_l - \mathbf{U}\Phi_{\theta^{(k+1)}(\mathbf{U})}(f_l)\|_2^2. \end{cases}$$

In each iteration, the updating step can be precisely expressed as the $\mathbf{V}$ block and the $\mathbf{U}$ block.

C. V Block

Here, the $\mathbf{V}$ block mainly updates the neural network Φ_θ when $\mathbf{U}$ is fixed. The input of Φ_θ corresponds to the wavelength in the lth band, while the output yields the corresponding endmember $\mathbf{v}_l \in \mathbb{R}^{J\times 1}$. In practice, we use the band index at each position as a substitute for wavelength if we have full uniform sampling covering all the available bands. For simplicity, the band index at the lth position is denoted as f_l in the following content. The endmember function is parameterized by a neural network θ, resulting in $\mathbf{v}_l = \Phi_\theta(f_l)$. Consequently, the process of endmember encoding involves adapting a neural network Φ_θ to given a spectral band index.

TABLE I
MLP ARCHITECTURE FOR $J \times L$ ENDMEMBER, CHANGE THE VALUE C AND J TO GET THE MODELS WITH DIFFERENT SIZES

Layer	Modules	Output Size
0	Positional Encoding	$2p \times 1$
1	MLPs	$C \times 1$
2	Head Layer	$J \times 1$

Essentially, the $\mathbf{V}$ block is a type of INR. It effectively creates a continuous representation that maps each input spectral band index into a corresponding endmember representation. INR can represent the corresponding endmembers well for any nonuniform sampling wavelength with a spectral domain.

a) Input embedding: The unmixing task's crucial aspect is acquiring a well-learned representation of the endmember details. Directly providing the network with coordinates or timestamps leads to undesirable results, as observed by previous studies [27], [40]. However, by mapping the inputs to a higher-dimensional embedding space, the neural network can more effectively adapt to data with high-frequency representation [41], [42]. In this regard, Positional Encoding [27], can enhance the network's ability to capture and represent complex patterns within the data. Specifically, in our method, we use $\tau(\cdot)$ as our embedding function, τ is a mapping from $\mathbb{R}$ into a higher dimensional space $\mathbb{R}^{2p}$ before passing them to the MLPs, and is defined as follows:

$$\tau(f_l) = \big[\tau_0(f_l), \tau_1(f_l), \ldots, \tau_{p-1}(f_l)\big], \ l = 1, \ldots, L \tag{6}$$

where $\tau_i(f_l) = (\sin(b^i \pi f_l), \cos(b^i \pi f_l))$ is a 2-D vector with b and p are hyper-parameters of the networks [27], b^i represents the sinusoidal coding of the signal at different frequencies at different times. Given an input band index f_l, normalized between (0, 1], the output of embedding function $\tau(\cdot)$ is then fed to the following neural network.

b) Network architecture: Our network architecture details are outlined in Table I. Here, the band index embedding serves as the input, and it undergoes a four-layer MLPs before generating the corresponding endmembers as the output. To accommodate various unmixing tasks, we train models of different sizes by adjusting the values of C and J.

c) Loss function: Within the NMF framework, additional constraints are required beyond the nonnegative constraint to ensure the problem is well-posed. Specifically, we calculate the endmember distance (ED) [43], [44] as the regularization $\mathcal{R}$ by measuring the distance between each endmember and its centroid. This measurement plays a crucial role in preserving the compactness of the endmembers and is closely linked to the minimum volume constraint (MVC) [43], which is to find a simplex with minimum volume. It is worth noting that the metric ED is equivalent to the endmember variance in which we compute the endmember variance for each band corresponding to the band location. We adopt a combination of ℓ_1 loss and endmember variance as our loss function for

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network optimization in the kth iteration as follows:

$$\frac{1}{L}\sum_{f_l\in\Omega}\left\{\|\mathbf{x}_l-\mathbf{U}^{(k)}\Phi_{\theta^{(k+1)}}(f_l)\|_1+\lambda \mathrm{ED}(\mathbf{v}_l)\right\} \tag{7}$$

where $\mathbf{U}^{(k)}\Phi_{\theta^{(k+1)}}(f_l)\in\mathbb{R}^{nm}$ is the prediction, $\mathbf{x}_l\in\mathbb{R}^{nm}$ corresponds to the HSI observation of the lth band, and

$$\mathrm{ED}(\mathbf{v}_l)=\|\mathbf{v}_l-\frac{1}{J}\sum_{j=1}^{J}v_{jl}\mathbf{1}_J\|_2^2$$

with $\mathbf{v}_l=[v_{1l},\ldots,v_{Kl}]^T$, λ is a regularization parameter to balance the regularization and data-fidelity term for each loss component. Note that our experiments show that the ℓ_1 norm as the data-fidelity term in (7) performs better than the ℓ_2 norm, see Section IV.

D. U Block

In the U block of the kth iteration, $\mathbf{V}^{(k+1)}$ is given and fixed. The optimization problem (5) turns to

$$\min_{\mathbf{U}\geq\mathbf{0}}\quad\|\mathbf{X}-\mathbf{U}\mathbf{V}^{(k+1)}\|_F^2. \tag{8}$$

When updating the abundance matrix $\mathbf{U}$, it's essential to maintain the ASC condition, which requires the normalization of the abundance. Instead of adding ASC as a constraint, we redefine the observation and spectral signature matrices, i.e., $\mathbf{X}$ and $\mathbf{V}^{(k+1)}$ in (8), as follows:

$$\mathbf{X}_\delta=\begin{bmatrix}\mathbf{X}&\delta\mathbf{1}_N\end{bmatrix},\quad \mathbf{V}_\delta^{(k+1)}=\begin{bmatrix}\mathbf{V}^{(k+1)}&\delta\mathbf{1}_J\end{bmatrix}.$$

This is equivalent to solve

$$\min_{\mathbf{U}\geq\mathbf{0}}\quad\|\mathbf{X}-\mathbf{U}\mathbf{V}^{(k+1)}\|_F^2+\delta^2\|\mathbf{U}\mathbf{1}_J-\mathbf{1}_N\|_2^2. \tag{9}$$

Here δ is a parameter that governs the influence of the ASC. This problem does not have a closed-form solution owing to the nonnegative constraint. Here the multiplicative update rule in NMF [36] is adapted to update $\mathbf{U}$ as

$$\mathbf{U}^{(k+1)}=\frac{\mathbf{U}^{(k)}\odot\left(\mathbf{X}_\delta\mathbf{V}_\delta^{(k+1)T}\right)}{\mathbf{U}^{(k)}\mathbf{V}_\delta^{(k+1)}\mathbf{V}_\delta^{(k+1)T}} \tag{10}$$

where the division is an element-wise operation, while the symbols $\odot$ represent element-wise multiplication.

Note that the training data in this framework only contains the matrix $\mathbf{X}$ and the corresponding bands index $\{f_l\}_{l=1}^L$. In a manner akin to NMF, the initialization of the endmember matrix, denoted as $\mathbf{V}$, employs the VCA method [37]. Simultaneously, the abundance matrix $\mathbf{U}$ is initialized using the fully FCLSs algorithm [38]. The whole procedure of the proposed NMFF algorithm is summarized as Algorithm 1.

Algorithm 1 NMFF Training Procedure.

Input: Observed data $\mathbf{X}$ and band index $\{f_l\}_{l=1}^L$
Output: Abundance map $\mathbf{U}$ and endmember $\mathbf{V}$

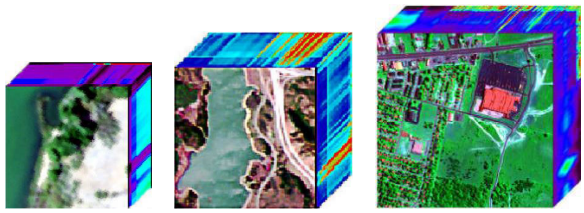
1: Initialize $\mathbf{U}$ and $\mathbf{V}$ by FCLS and VCA, augment $\mathbf{X}$ into $\mathbf{X}_\delta$, choose parameter λ and δ
2: **repeat**
3: Update $\theta^{(k+1)}$ by training the neural network
4: Update endmember

$$\mathbf{V}^{(k+1)}=[\Phi_{\theta^{(k+1)}}(f_1),\ldots,\Phi_{\theta^{(k+1)}}(f_L)]$$

5: Augment $\mathbf{V}^{(k+1)}$ into $\mathbf{V}_\delta^{(k+1)}$
6: Update abundance $\mathbf{U}^{(k+1)}$ by (10)
7: **until** convergence

IV. Experiments Results

In this section, we provide experimental results evaluating the effectiveness of the proposed approach with various datasets. Our method (NMFF) is compared with several state-of-the-art spectral blind unmixing approaches, including NMF-based approaches ($L_{1/2}$-NMF [12], MLNMF [13], and NMF-QMV [45]), as well as deep learning methods (UnDIP [22], TRA-NET [23] and DGMCNN [24]). All numerical experiments about these NMF-based methods are conducted using the MATLAB R2022a with an Intel Core i7-10870H CPU with 3.10-GHz, 16-GB RAM. Other deep learning-based algorithms are run on the NVIDIA A100-SXM4 GPU with 40 GB memory.



(a) (b) (c)

Fig. 3. Visualization of (a) Samson image, (b) Jasper Ridge image, and (c) Urban image.

TABLE II
Hyperparameters Used for Training the Proposed Model

Hyperparameters	Samson	Jasper	Urban
λ	0.6	1.5	0.35 or 0.22
δ	15	1.8	10
b	1.25	1.25	1.25
p	15	15	15
Epoch	30	100	30
Learning rate	0.0005	0.0005	0.0001

A. Data Description and Experimental Settings

1) Samson: As shown in Fig. 3(a), the Samson hyperspectral dataset contains 95×95 pixels, and each pixel has 156 bands ranging from 401 to 889 nm. This dataset sets the number of endmembers to 3, including Soil, Tree, and Water. The GT endmember spectra[1] were manually selected from the image.
2) Jasper Ridge: As shown in Fig. 3(b), this dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). it contains 100×100 pixels, and each pixel has 198 bands (i.e., 4–107, 113–153, and 167–219) after removal of noisy bands ranging from 380 to

[1]https://rslab.ut.ac.ir/data

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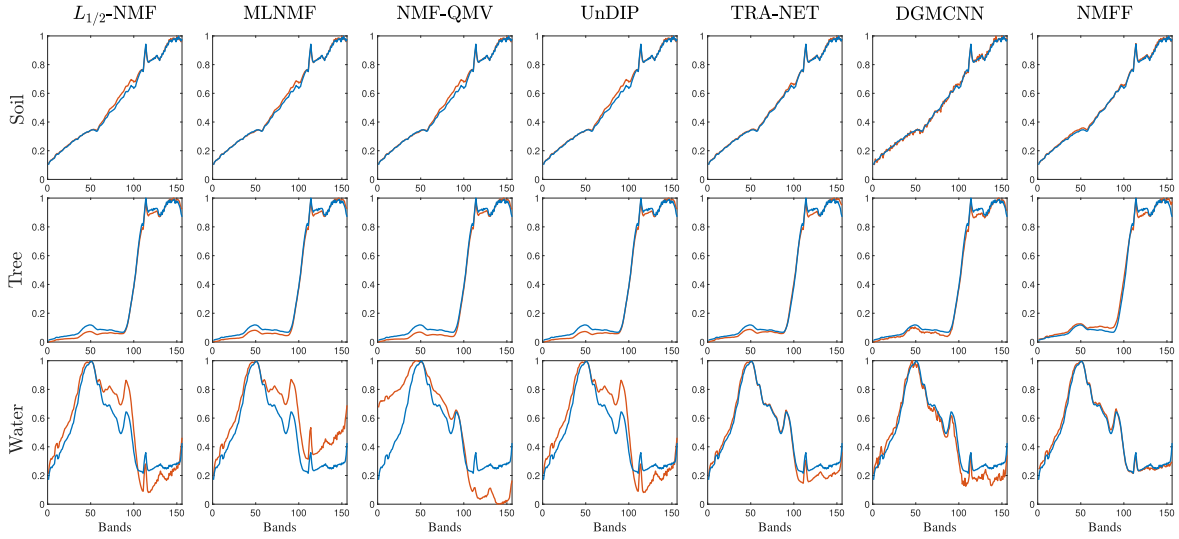


Fig. 4. Estimated endmembers comparison for different methods on the Samson dataset under uniform frequency sampling. (Blue) GT endmembers. (Red) Estimated endmembers.

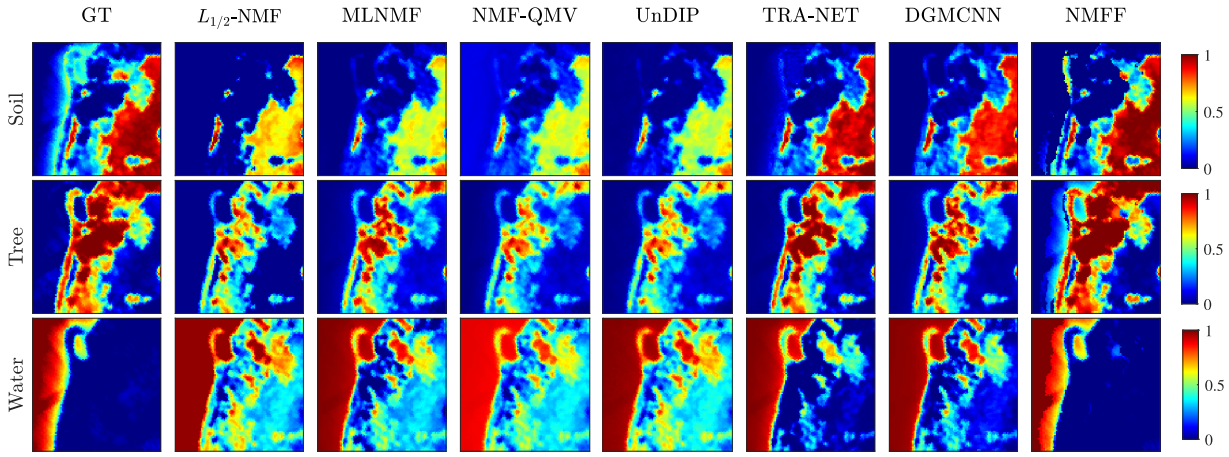


Fig. 5. Abundance maps of unmixing results for different methods on the Samson dataset under uniform frequency sampling.

2500 nm. The number of endmember spectra is set to 4, including Tree, Water, Soil, and Road.

3) Urban: As shown in Fig. 3(c), Urban is one of the most widely used hyperspectral data in the hyperspectral unmixing study. There are 307 × 307 pixels, each corresponding to a 2 × 2 m^2 area. This image has 210 wavelengths ranging from 400 to 2500 nm. After removing channels 1–4, 76, 87, 101–111, 136–153, and 198–210 (due to dense water vapor and atmospheric effects), we remain 162 channels. The number of endmember spectra is set to 4, including Asphalt, Grass, Tree, and Roof.

1) Performance Metrics: We use the root mean square error (RMSE) for the abundances to evaluate the performance of the comparison models. The RMSE metric for the j-endmember is defined as

$$\text{RMSE}_j = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|\bar{u}_{ij} - u_{ij}\|_2^2}$$

where u_{ij} and $\bar{u}_{ij}$ are the ith entry of the GT and estimated abundance vector for the jth endmember, respectively. The spectral angle distance (SAD) is introduced to measure the similarity between the GT endmembers $\mathbf{V}$ and the estimates $\hat{\mathbf{V}}$, as defined by

$$\text{SAD}_j = \arccos\left(\frac{\mathbf{V}_j^T\hat{\mathbf{V}}_j}{\|\mathbf{V}_j\|_2\|\hat{\mathbf{V}}_j\|_2}\right)$$

where $\mathbf{V}_j$ and $\hat{\mathbf{V}}_j$ are the jth row of $\mathbf{V}$ and $\hat{\mathbf{V}}$, respectively. The overall performances of estimating both abundances and endmembers are evaluated by their average values, respectively,

$$\text{aRMSE} = \frac{1}{J}\sum_{j=1}^{J}\text{RMSE}_j, \quad \text{aSAD} = \frac{1}{J}\sum_{j=1}^{J}\text{SAD}_j.$$

2) Hyperparameter Settings: The proposed method utilizes specific hyperparameters as described below: The INR network is structured with four layers, each having a width of 256. This network's initialization method and activation function aligns with SIREN [28]. The initial parameter ω_0 in SIREN for each MLP layer is uniformly set to 5 for all experiments. The training of endmembers is carried out using the Adam optimizer. Table II presents the selected hyperparameters for training the proposed model. It's important to note that the

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TABLE III

aSAD (Endmember) and aRMSE (Abundance) Scores (Average of Ten Runs) Along With Their Standard Deviation With Samson Data Under Uniform Frequency Sampling

Method	$L_{1/2}$-NMF	MLNMF	NMF-QMV	UnDIP	TRA-NET	DGMCNN	NMFF
Soil	0.0268±0.0100	0.0185±0.0002	0.0233±0.0001	0.0404±0.0009	0.0249±0.0109	0.0235±0.0028	**0.0127±0.0003**
Tree	0.0409±0.0016	0.0422±0.0005	0.0515±0.0001	**0.0219±0.0002**	0.0464±0.0066	0.0361±0.0001	0.0490±0.0001
Water	0.4006±0.3964	0.1735±0.0117	0.3377±0.0011	0.1304±0.0124	0.0943±0.0192	0.0976±0.0032	**0.0382±0.0024**
aSAD	0.1561±0.1339	0.0780±0.0040	0.1375±0.0003	0.0642±0.0042	0.0552±0.0114	0.0524±0.0013	**0.0333±0.0001**
soil	0.2458±0.0208	0.2542±0.0006	0.2333±0.0001	0.2699±0.0011	0.1618±0.0110	0.1878±0.0042	**0.1408±0.0001**
tree	0.2653±0.0231	0.1955±0.0066	0.2442±0.0001	0.2516±0.0031	0.1578±0.0332	0.1806±0.0079	**0.1046±0.0001**
water	0.4389±0.0119	0.3624±0.0071	0.3763±0.0005	0.4260±0.0041	0.2672±0.0318	0.3035±0.0073	**0.0454±0.0019**
aRMSE	0.3167±0.0084	0.2707±0.0048	0.2845±0.0002	0.3149±0.0028	0.2022±0.0256	0.2240±0.0058	**0.0969±0.0001**

TABLE IV

aSAD (Endmember) and aRMSE (Abundance) Scores With Samson Data Under Nonuniform Frequency Sampling

Method	$L_{1/2}$-NMF	MLNMF	NMF-QMV	UnDIP	TRA-NET	DGMCNN	NMFF
soil	0.0253	0.0171	0.0427	0.0441	0.0210	0.0188	**0.0120**
tree	0.0432	0.0456	0.0573	**0.0231**	0.0489	0.0381	0.0406
water	0.1567	0.1405	0.3729	0.1390	0.0897	0.0985	**0.0738**
aSAD	0.0749	0.0677	0.1519	0.0687	0.0532	0.0518	**0.0421**
soil	0.2134	0.2516	0.2332	0.2666	0.1592	0.1725	**0.1410**
tree	0.3115	0.1782	0.2461	0.2516	0.1492	0.1838	**0.0863**
water	0.4305	0.3466	0.3769	0.4255	0.2601	0.2832	**0.1076**
aRMSE	0.3185	0.2588	0.2854	0.3146	0.1895	0.2132	**0.1116**

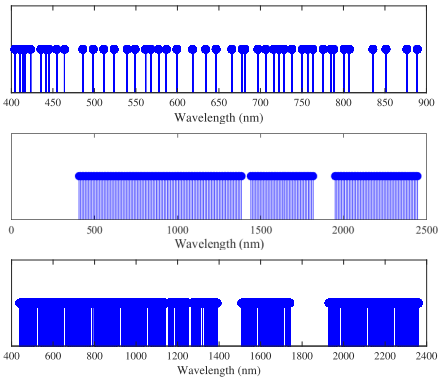


Fig. 6. Nonuniform wavelength sampling in (top) Samson, (middle) Jasper Ridge, and (bottom) Urban datasets.

hyperparameters for the nonuniform Samson data are the same as those defined for the Samson dataset, hence, they are not listed in Table II.

B. Experimental Results on the Uniform Frequency Sampling With Samson Dataset

The frequencies of the Samson data follow a uniform distribution. The performance comparison in terms of aRMSE for abundances and aSAD for endmembers for all methods are shown in Table III, while the corresponding endmembers and abundance maps are illustrated in Figs. 4 and 5, respectively. Some of the comparison algorithms struggle to completely distinguish the Water and Tree abundance maps due to the challenge of extracting certain endmembers from the VCA initialization. Although deep learning methods outperform traditional NMF algorithms, there is still considerable room for improvement in estimating abundance maps and endmembers, as indicated in Table III. NMFF's estimated endmember signatures align well with the reference GT, demonstrating its effectiveness for unmixing in real-world scenarios. Additionally, as shown in Fig. 5, the estimated abundance maps generated by NMFF closely match GT.

To validate the stability of various methods, we repeated the experiment ten times, obtaining aSAD and aRMSE scores, along with their standard deviations. As shown in Table III, it is evident that the stability of our method is significantly superior to other deep learning approaches. It is noteworthy that, to save computational time, we conducted only a single experiment in the following trials.

C. Experimental Results on Nonuniform Frequency Sampling With Samson Dataset

To assess the performance of NMFF on a nonuniformly sampled dataset derived from the Samson data, which originally comprised 156 frequencies as depicted in Fig. 6(a), we implemented a nonuniform sampling process, resulting in 45 nonuniform wavelengths. For this nonuniform wavelength dataset, we present the quantitative results of various comparative methods in Table IV, alongside the corresponding endmembers and abundance maps showcased in Figs. 7 and 8, respectively. The results in Table IV reveal that certain methods, such as $L_{1/2}$-NMF and NMF-QMV, exhibit subpar

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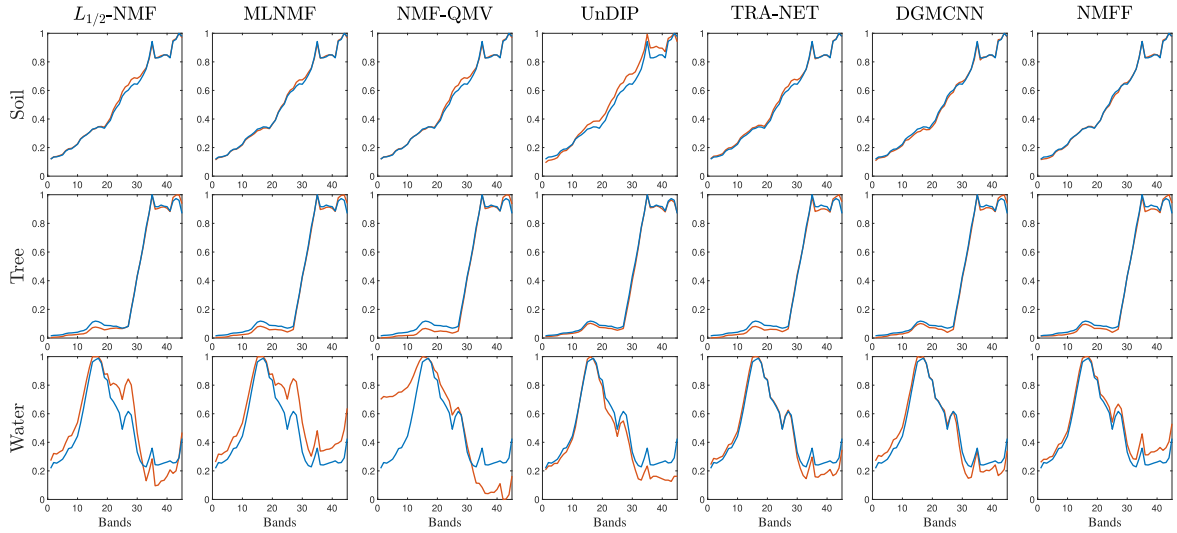


Fig. 7. Estimated endmembers comparison for different methods on the Samson dataset with nonuniform frequency. (Blue) GT endmembers. (Red) Estimated endmembers. The missing bands are skipped.

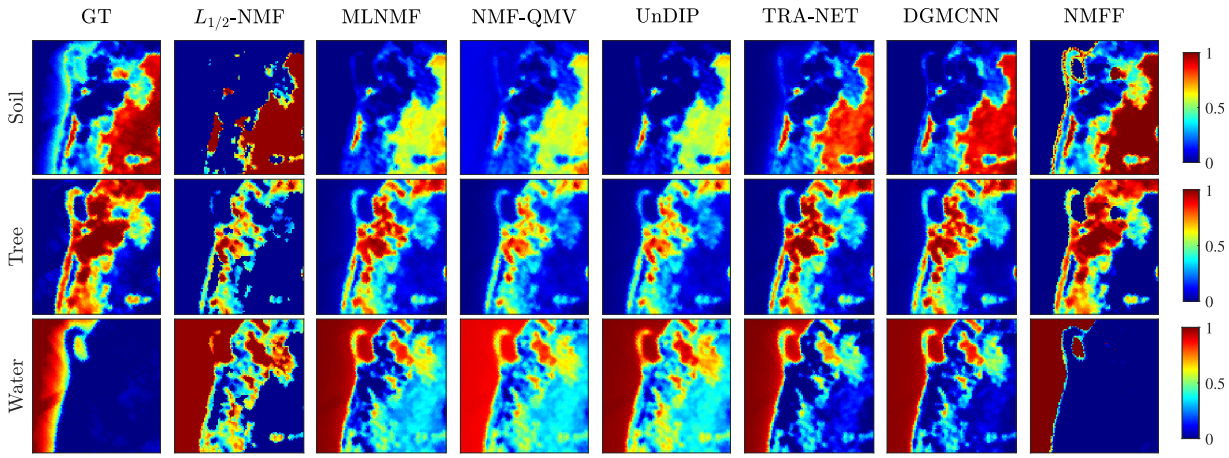


Fig. 8. Abundance maps of unmixing results for different methods on the Samson dataset with nonuniform frequency.

TABLE V
ASAD SCORES (ENDMEMBER) AND ARMSE (ABUNDANCE) WITH JASPER RIDGE DATA

Method	$L_{1/2}$-NMF	MLNMF	NMF-QMV	UnDIP	TRA-NET	DGMCNN	NMFF
Tree	0.1657	0.1456	0.2726	0.1070	0.0862	0.1124	**0.0533**
Water	0.2532	0.1421	0.4160	0.2532	**0.0997**	0.2780	0.1983
Soil	0.1510	0.1576	0.1723	0.1289	0.1870	0.0434	**0.0397**
Road	0.2777	0.6736	0.0498	0.0901	0.1047	**0.0462**	0.0527
aSAD	0.2023	0.2797	0.2277	0.1448	0.1194	0.1200	**0.0861**
Tree	0.1156	0.1774	0.2044	0.1565	0.1848	0.0955	**0.0426**
Water	0.2358	0.1788	0.1840	0.2205	0.0902	0.0864	**0.0409**
Soil	0.2446	0.1739	0.1464	0.1301	0.2306	0.1170	**0.0669**
Road	0.2087	0.1737	0.1246	0.1088	0.1958	0.0700	**0.0567**
aRMSE	0.2012	0.1760	0.1649	0.1540	0.1753	0.0922	**0.0551**

performance compared to the results obtained with uniform sampling, as presented in Table III. In contrast, NMFF excels in accurately estimating the abundance maps and endmembers, showcasing an approximately 57.9% improvement in aRMSE over the DGMCNN method.

The estimated endmember signatures, illustrated in Fig. 7, further demonstrate the effectiveness of our proposed model. Additionally, the abundance map displayed in Fig. 8 highlights NMFF's ability to differentiate between various types of materials effectively. NMFF demonstrates robustness and

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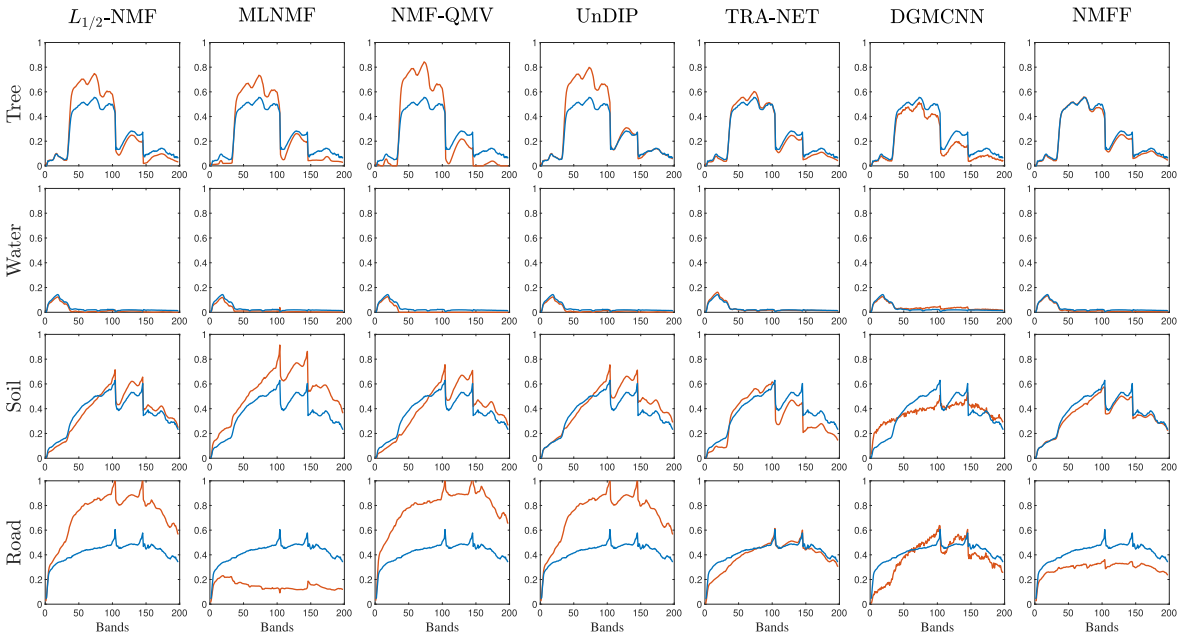


Fig. 9. Estimated endmembers comparison for different methods on the Jasper Ridge dataset. (Blue) GT endmembers. (Red) Estimated endmembers. The missing bands are skipped.

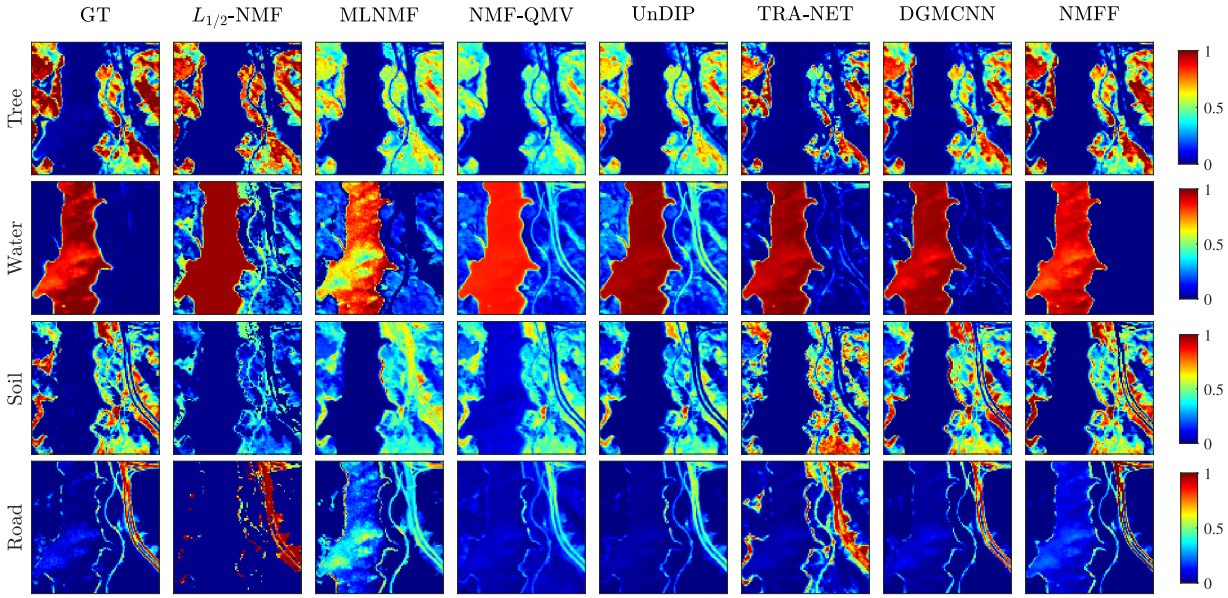


Fig. 10. Abundance maps of unmixing results for different methods on the Jasper Ridge dataset.

efficacy even when applied to datasets with nonuniform wavelengths, outperforming certain comparison methods that might encounter challenges with such datasets. Its consistent delivery of accurate abundance maps and endmember estimations renders it a suitable and dependable choice for a range of real-world scenarios.

D. Experimental Results on the Nonuniform Frequency Sampling With Jasper Ridge Dataset

Table V presents the quantitative results obtained from the Jasper Ridge dataset. It is evident that the endmember labeled as Road presents a considerable challenge for the other methods, but NMFF successfully achieves a satisfactory estimation for this particular endmember. The proposed method outperforms all other unmixing techniques, achieving an aRMSE value of 0.0551 and an aSAD value of 0.0861.

In Fig. 9, NMFF demonstrates the most accurate endmember estimation for Tree and Soil regarding aSAD. While the endmember for Water shows a relatively poorer estimation, it still manages to capture the general shape of the endmember accurately. On the other hand, Fig. 10 shows that specific comparison algorithms struggle to completely separate the Road abundance map due to difficulties in extracting certain endmembers.

E. Experimental Results on the Nonuniform Frequency Sampling With Urban Dataset

We conducted further testing of our proposed model on the Urban dataset, which features nonuniformly distributed

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TABLE VI
ASAD SCORES (ENDMEMBER) AND ARMSE (ABUNDANCE) WITH URBAN DATA

Method	$L_{1/2}$-NMF	MLNMF	NMF-QMV	UnDIP	TRA-NET	DGMCNN	NMFF
Asphalt	0.2139	0.0896	0.3076	0.1698	0.2967	**0.0648**	0.0934
Grass	0.1992	0.3132	0.1454	0.2836	0.1768	0.0616	**0.0441**
Tree	0.1549	0.0971	0.1128	0.1548	0.1346	0.0772	**0.0463**
Roof	0.7705	0.4911	0.0871	0.0778	0.2686	**0.0296**	0.0454
aSAD	0.3346	0.3303	0.1632	0.1715	0.2192	0.0583	**0.0573**
Asphalt	0.3403	0.3385	0.2658	0.2304	0.1434	0.3513	**0.0892**
Grass	0.4756	0.3963	0.3181	0.2861	0.1776	0.2253	**0.1043**
Tree	0.5241	0.4626	0.3704	0.1906	0.1127	0.1984	**0.0663**
Roof	0.1928	0.1350	0.1997	0.2288	0.1295	0.3152	**0.0685**
aRMSE	0.4043	0.3549	0.2954	0.2365	0.1428	0.2797	**0.0821**

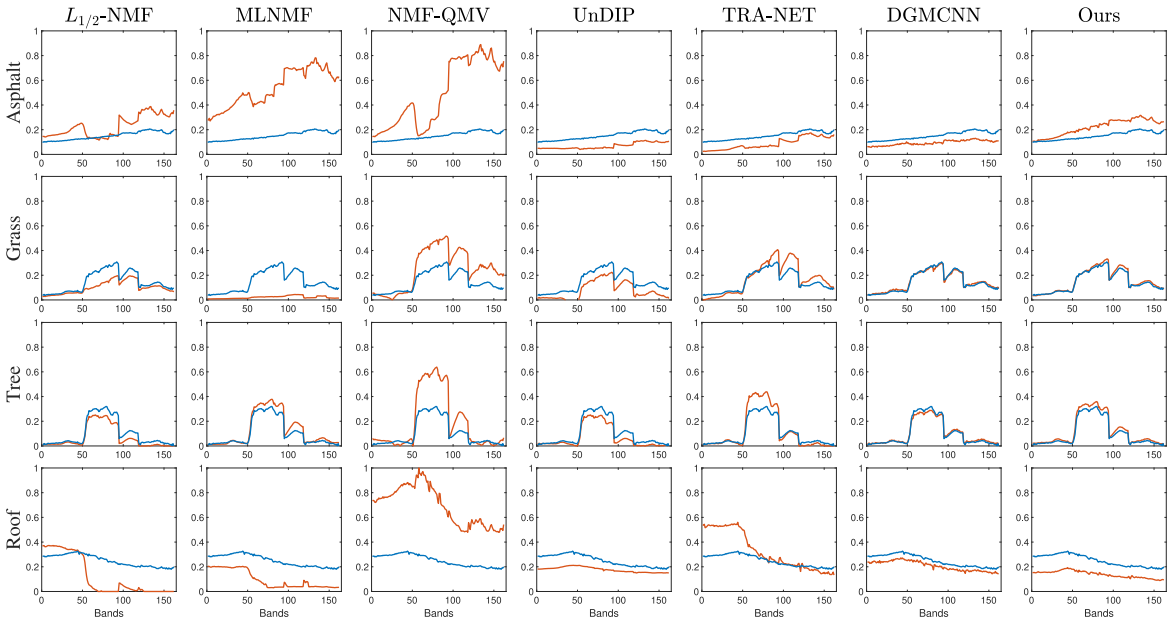


Fig. 11. Estimated endmembers comparison for different methods on the Urban dataset. (Blue) GT endmembers. (Red) Estimated endmembers. The missing bands are skipped.

TABLE VII
ACTIVATION FUNCTION ABLATION

	aSAD	aRMSE
ReLU	0.0568	0.0825
Leaky ReLU	0.0570	0.0831
Sin	0.0573	**0.0821**
Swish	**0.0564**	0.0823
WIRE	0.1165	0.0954

TABLE VIII
MLP LAYERS ABLATION

	aSAD	aRMSE
3	0.0580	0.0824
4	**0.0573**	**0.0821**
5	0.0587	0.0820
6	0.0594	0.0827

wavelengths [as shown in Fig. 6(c)]. The results of abundance estimations are presented in Table VI, and the visual comparisons of abundance maps and endmembers are shown in Figs. 11 and 12, respectively. NMFF achieves the lowest aSAD among all approaches and excels in estimating the endmember for Grass and Tree. On the other hand, the DGMCNN unmixing method performs best in estimating Asphalt and Roof endmembers. Regarding the overall aRMSE, NMFF achieves the lowest value and produces the most accurate abundance estimations for all materials.

The visual comparison in Figs. 11 and 12 confirms the findings reported in Table VI. The estimated endmember signatures in Fig. 11 further highlight the effectiveness of our proposed model. NMFF exhibits superior accuracy in

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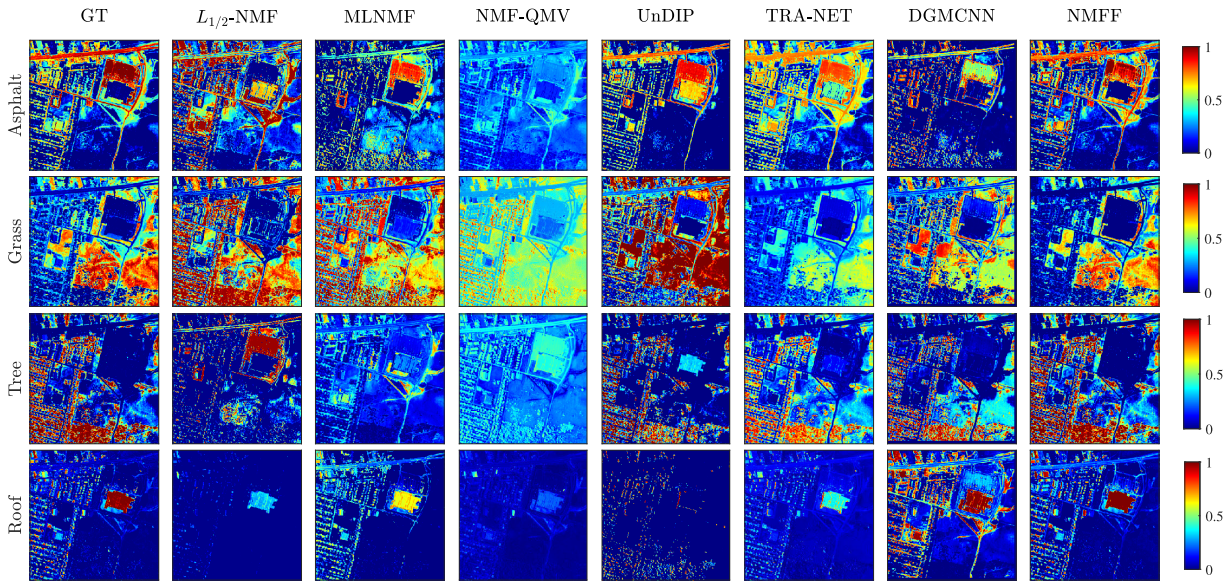


Fig. 12. Abundance maps of unmixing results for different methods on the Urban dataset.

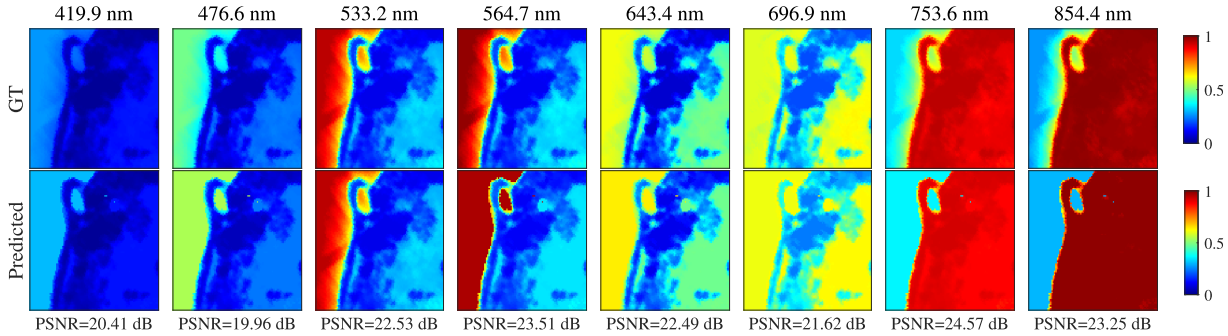


Fig. 13. Restoration of some missing hyperspectral images at 419.9, 476.6, 533.2, 546.7, 643.4, 696.9, 753.6, and 854.4 nm on the Samson dataset.

TABLE IX
LOSS FUNCTION ABLATION

ℓ_2	ℓ_1	ED	aSAD	aRMSE
✓			0.1741	0.1069
	✓		0.2463	0.1164
✓		✓	0.1438	0.1430
	✓	✓	**0.0573**	**0.0821**

abundance estimation, particularly in distinguishing similar materials, as illustrated in Fig. 12 for Grass and Tree.

F. Hyperspectral Super-Resolution

Since our approach can learn a continuous representation of the endmember, we utilize this representation and explore the hyperspectral super-resolution as an additional reward from our proposed method. Specifically, after we train our model Φ_θ with a subset of bands sampled from the Samson dataset, we use the trained model to predict mixing bands given a missing frame index, i.e.,

$$\mathbf{x}_{f_i} = \mathbf{U}\Phi_\theta(f_i), \quad \text{for all } f_i \in \Omega^C.$$

As shown in Fig. 13, our method can give quite reasonable predictions on the missing bands, which have good and comparable visual quality compared to GT. Therefore, all of these results validate that our method can effectively predict the HSI data corresponding to those missing spectral wavelengths.

G. Ablation Study

In this section, we perform a series of ablation studies on our proposed method to comprehensively analyze the influence of various factors on the parametric model of INR. Through these comparative studies on the Urban dataset, we aim to gain valuable insights into these impacts on the model's overall performance. To evaluate the quality of our reconstructions, we employ two standard metrics, namely aSAD and aRMSE, to assess the accuracy of the endmembers and abundance maps, respectively.

1) Effect of Activation Layer: Table VII presents the outcomes achieved with various activation functions, including ReLU, Leaky ReLU, Sin [28], Swish, and WIRE [29]. Remarkably, the Sin activation function showcasing exceptional performance across the board. Consequently, we have chosen it as the default activation function for our design.

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TABLE X
COMPARISON OF THE COMPUTATIONAL TIME WITH URBAN DATASET (UNITS: SECONDS)

Method	$L_{1/2}$-NMF	MLNMF	NMF-QMV	UnDIP	TRA-NET	DGMCNN	NMFF
Time	144.38	**32.22**	151.59	662.58	285.11	289.13	420.08

2) Effect of Number of Layers: In Table VIII, we show the results with different numbers of layers. With four layers, the default setup reaches the best performance and runs slightly faster.

3) Effect of Loss Function in the **V** *Block:* We show the ablation study on the loss function of the **V** block in Table IX and test different combinations of the ℓ_2, ℓ_1, for the data-fidelity term and the ED regularization. The combination of the ℓ_1 data fidelity and the ED regularization can achieve the best trade-off between the aSAD and aRMSE performance.

4) Running Time: As shown in Table X, the running time of model-based methods is faster than deep learning methods. Specifically, our proposed method is faster than UnDIP but slower than TRA-NET and DGMCNN due to the presence of both inner and outer loops in our training process.

V. CONCLUSION

In this article, we introduce an innovative unmixing model that combines INR and NMF to enhance HSI unmixing effectiveness. The key sight of our model involves training a neural network directly with spectral band index, generating corresponding endmember values as outputs. By employing a continuous representation function, we can effectively express endmembers, thus capturing the nonuniform characteristics of spectral wavelengths. This unique endmember representation allows us to apply the iterative approach of NMF for abundance extraction, eliminating the necessity for a complex and lengthy processing pipeline. The results of our unmixing technique on various datasets, including Samson, Ridge, and Urban, clearly demonstrate the improved performance of our proposed model over existing methods. In our future work, a significant focus will be directed toward enhancing hyperspectral unmixing performance by incorporating regularization terms into the abundance matrix **U**. Moreover, the proposed continuous representation demonstrates significant potential as a versatile tool for HSI analysis. It holds promise for further extension into emerging tasks such as HSI fusion and super-resolution, presenting exciting opportunities for advancing the field. Our proposed NMFF has certain limitations. The inclusion of both inner and outer loops in our training process leads to increased computational and time costs. Furthermore, there is a lack of a rigorous theoretical analysis for the proposed method. In our future directions, we will prioritize conducting a thorough theoretical analysis and explore further advancements in endmember representation to enhance unmixing performance.

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